Total No. of Printed Pages: 03

SUBJECT CODE NO: - RNP-8001
FACULTY OF SCIENCE AND TECHNOLOGY
F. E. (NEP) [Group A & B] Mech, Civil, Comp. AI & ML, EEE
Examination (January/February-2025)
Applied Mathematics - I

[Time: 3:00 Hours] **[Max. Marks: 60]**

Please check whether you have got the right question paper.

N. B
1) Use of non-programmable calculator is allowed
2) Q.no.1 is compulsory
3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Solve the following questions: (Any 10) **20**

a) Which of the following functions can be expressed as a Maclaurin's series?
   a) Any differentiable function.
   b) Functions that are differentiable at x=0.
   c) Only polynomial functions.
   d) Any continuous

b) Taylor's theorem is valid for which types of functions?
   a) Continuous functions
   b) Functions that is infinitely differentiable.
   c) Only polynomial functions.
   d) Functions with a finite number of derivatives question/multiple

c) When evaluating $\lim_{x\to\infty} \frac{x}{e^x}$ which indeterminate form arises?
   a) 0/0   b) ∞ / ∞
   c) $0 \times \infty$ d) 1^0

d) By the Cauchys nth root test the series $\sum_{n=1}^{\infty} u_n$ is convergent if
   a) $\lim_{n\to\infty}(u_n)^{\frac{1}{n}} < 1$ b) $\lim_{n\to\infty}(u_n)^{\frac{1}{n}} > 1$
   c) $\lim_{n\to\infty}(u_n)^{\frac{1}{n}} = 1$ d) 0

e) The maximum value of the function sinx.siny.sin(x+y) is
   a) $\frac{3}{2}$ b) $\frac{\sqrt{3}}{2}$
   c) $\frac{3\sqrt{3}}{8}$ d) 1

f) The Integrating Factor of $\frac{dy}{dx} + xy = x^3$D.E is

a) $x^2y = c$ b) $x = c$

c) $e^{\frac{x^2}{2}}$ d) $x^2y - xy = c$

g) The necessary & sufficient condition for D.E. to be exact D.E. is

a) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$ b) $\frac{\partial M}{\partial y} = -\frac{\partial N}{\partial x}$

c) $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$ d) $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$

h) A circuit containing resistance R and inductance L in series with voltage source E then the D.E for current is

a) $Li + R\frac{di}{dt} = E$ b) $L\frac{di}{dt} + Ri = E$

c) $Li - R\frac{di}{dt} = E$ d) $Li + R\frac{di}{dt} = 0$

i) If $u = \frac{x^2y^2z^2}{x^2+y^2+z^2}$ then $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z}$ is equal to

a) 0 b) 1

c) 4u d) u

j) If u, v & w are functions of three independent variables x, y & z then the Jacobean is given by

a) $J = \frac{\partial(x,y,z)}{\partial(u,v,w)}$ b) $J = \frac{\partial(x,v,z)}{\partial(u,y,w)}$

c) $J = \frac{\partial(u,v,w)}{\partial(x,y,z)}$ d) $J = \frac{\partial(z,y,x)}{\partial(w,y,x)}$

k) A body of mass m falls from rest under gravity, in a fluid whose resistance to motion at any time t is mk times its velocity where k is constant, the D.E. of motion is

a) $\frac{dv}{dt} = -g - kv$ b) $\frac{dv}{dt} = g - kv$

c) $\frac{dv}{dt} = -g + kv$ d) $\frac{dv}{dt} = mg - mkv$

l) If $u = \log(x^3 + y^3)$ then $\frac{\partial u}{\partial x}$ is

a) $\frac{3x+3y}{x^3y^3}$ b) $\frac{3x^2}{x^3+y^3}$

c) $\frac{3y}{x^3y^3}$ d) $\frac{3y^2}{x^3+y^3}$

m) Find $\frac{\partial z}{\partial y}$ where z = ax^2+2by^2+2bxy,
 a) 3by b) 2ax
 c) 3(ax+by) d) 4by+2bx

SECTION B

Q2 a) Use Taylors theorem to find $\sqrt{25.15}$ **05**

b) Prove that $\lim_{x\to 0}\left(\frac{1^x+2^x+3^x+4^x}{4}\right)^{\frac{1}{x}} = (24)^{1/4}$ **05**

Q3 a) If $xu = yz, yv = zx, zw = xy$ then find $\frac{\partial(u,v,w)}{\partial(x,y,z)}$ **05**

b) If $u = \tan^{-1}\left(\frac{x^3+y^3}{x-y}\right)$, prove that $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = sin4u - sin2u$ **05**

Q4 a) Solve: $\sin x\frac{dy}{dx} + y\cos x = 2\sin^2 x.\cos x$ **05**

b) Solve: $\frac{dy}{dx} = \frac{\tan y-2xy-y}{x^2-xtan_y^2+\sec^2 y}$ **05**

Q5 a) Explain log(1+sinx) in powers of x. **05**

b) Test the convergence of $\sum_{n=1}^{\infty}\frac{1}{(n+1)^2}$ **05**

Q6 a) If $z = x^y$ then show that $\frac{\partial^2 z}{\partial x.\partial y} = \frac{\partial^2 z}{\partial y.\partial x}$ **05**

b) Divide 24 into three parts such that the continued product of the first square of the Second & cube of the third is maximum **05**

Q7 a) $3y^2\frac{dy}{dx} + 2xy^3 = 4xe^{-x^2}$ **05**

b) A body of mass m, falling from rest is subjected to the force of gravity & an air resistance proportional to the square of the velocity. If it falls through a distance x & possesses a velocity v at that instant, Prove that $\frac{2kx}{m} = \log(\frac{a^2}{a^2-v^2})$where $mg = ka^2$. **05**